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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SURFACE ROUGHNESS
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SYMBOL													

D

C

B

A

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1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 - 6	6 - 30	30 - 120	120 - 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

CN1

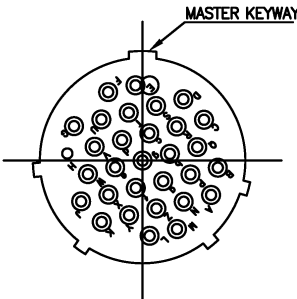
PIN NO	WIRE NO	TO	REMARK
A	0V620-1	EDB	
B			
C	0V704-1	EDB	
D	0V705-1	EDB	
E	0V720-1	EDB	
F	0V721-1	EDB	
G			
H	0V812-1	EDB	
J	0V853-1	EDB	
K	0V851-1	EDB	
L	0V942	AAU	
M	0V411	BECU T/B	
N	0V412	BECU T/B	
P	0V611-5	EDB	
Q	0V610-5	EDB	
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
f			
g			

CN2

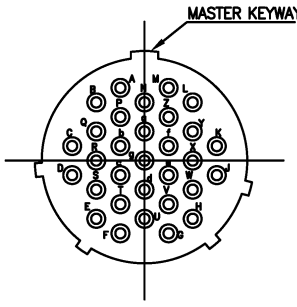
PIN NO	WIRE NO	TO	REMARK
A	0V212	EDB	
B			
C	0V215	EDB	
D	0V216	EDB	
E			
F	0V214	EDB	
G	0V231-1	EDB	
H	0V301	VWF INVERTER	
J	0V302	VWF INVERTER	
K			
L	0V811-1	EDB	
M	0V943-8	EDB	
N	0V971-1	ETHERNET SWITCH	
P			
Q			
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
f			
g			

CN3

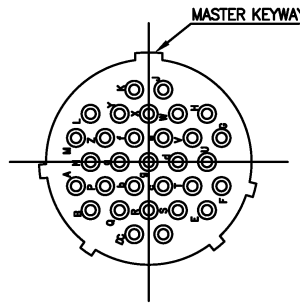
PIN NO	WIRE NO	TO	REMARK
A			
B			
C			
D			
E			
F			
G			
H			
J			
K			
L			
M			
N			
P			
Q			
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
f			
g			



CN1 : 24-31SX (31 #16)



CN2 : 24-31PN (31 #16)



CN3 : 24-31SW (31 #16)

- * NOTE
1. ALL CABLE EXCEPT SHIELD CABLE SHALL BE THE 3GKW-1.5 FOR CONNECTOR

SP153
SP154
SP155
SP105
SP106
SP107
SP138
SP139
SP140
SP149

						PRODUCT	KMRCL RS(3R) METRO CARS			
						REF DRG				
						MATERIAL				
						HEAT TREAT.	APPD	K.C.SHASHIKANTH	28.01.2018	
						SURFACE TREAT.	REVD	SADHASIVAM.M	28.01.2018	
						TITLE	CHKD	SARANYA.V	28.01.2018	
						PIN ASSIGNMENT, LTEB - M CAR	DRWN	SUSHILA PATIL	28.01.2018	
							SCALE		SHEET	Wt.(Kg)
							NTS		1 OF 2	0
							DRG No.			ALT
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEML LIMITED	942-38705			

D

C

B

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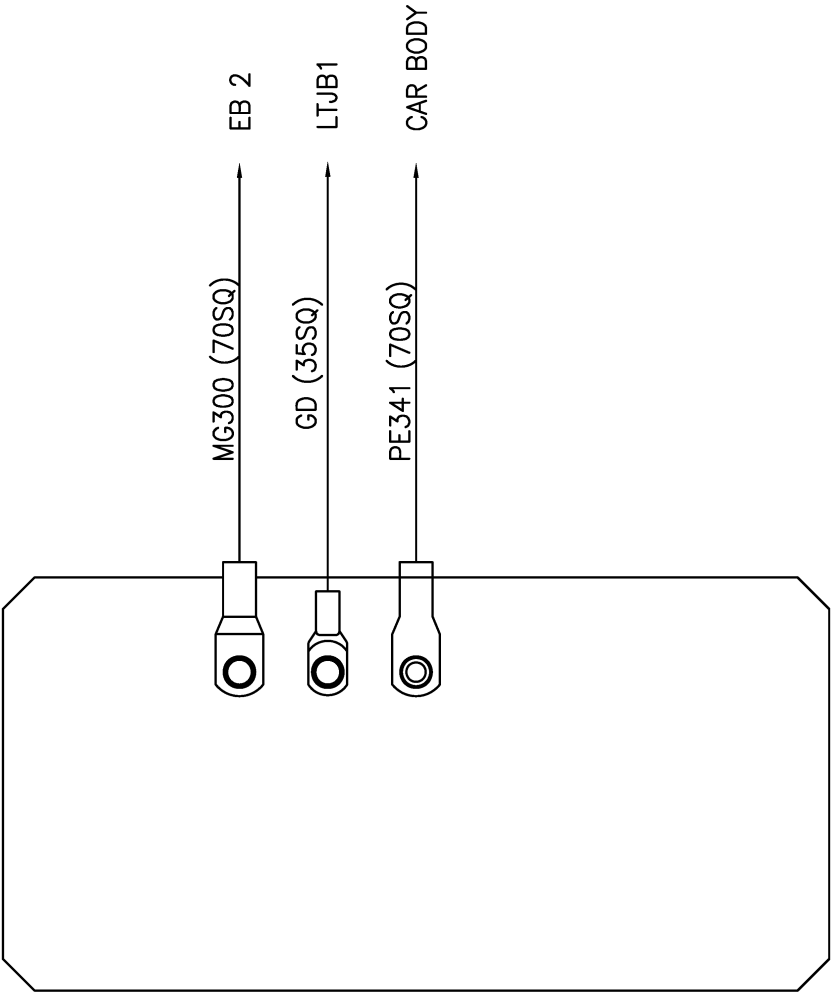
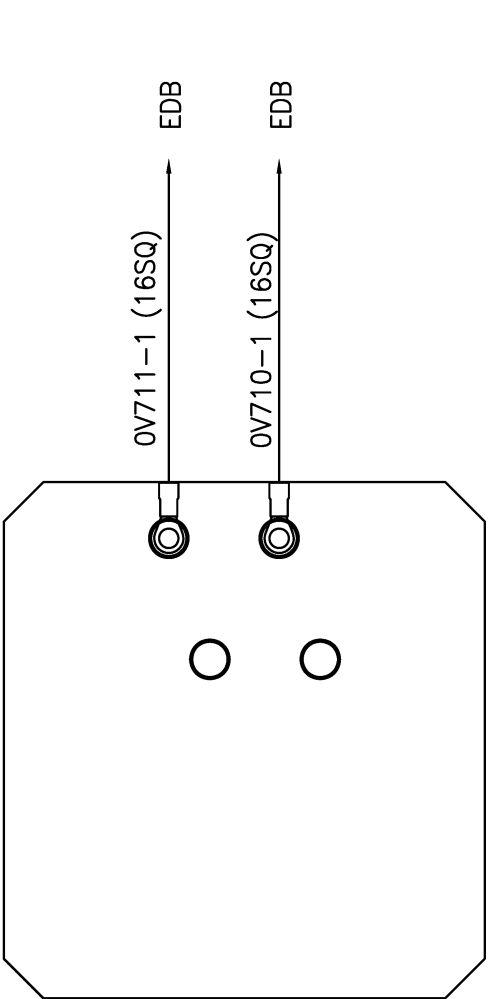
1

A3

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GRADE No.	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1
VALUE	50	25	12.5	6.3	3.2	1.6	0.8	0.4	0.2	0.1	0.05	0.025
SYMBOL												
SURFACE ROUGHNESS												

TERMINAL WIRING



MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADHASIVAM.M	27.01.2018	
						TITLE	PIN ASSIGNMENT, LTEB - M CAR				
							SCALE			SHEET	Wt.(Kg)
							NTS			2 OF 2	0
							DRG No.				ALT
							942-38705				
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKO	APPD	BEML LIMITED NEW FRONTIERS. NEW DREAMS					

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GRADE No.	VALUE	SYMBOL
N1	0.025	▽
N2	0.05	▽
N3	0.1	▽
N4	0.2	▽
N5	0.4	▽
N6	0.8	▽
N7	1.6	▽
N8	3.2	▽
N9	6.3	▽
N10	12.5	▽
N11	25	▽
N12	50	▽
SURFACE ROUGHNESS		

APPLICABLE FROM 4TH M CAR ONWARDS

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 - 6	6 - 30	30 - 120	120 - 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

CN1 (HL12)

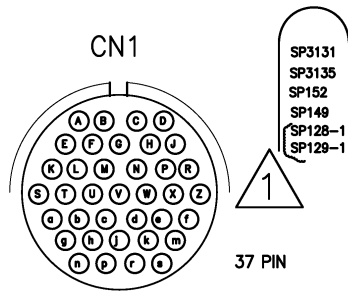
PIN NO	WIRE NO	TO	REMARK
A			
B	M3020-1	EDB	
C	M3020-2	EDB	
D			
E			
F	M3021-1	EDB	
G	0V301	LTEB	
H	0V302	LTEB	
J			
K	3010-1	EDB	
L	3016-1	EDB	
M	3011-1	EDB	
N	3005-1	EDB	
P	3006-1	EDB	
R	M3020a-2	BECU TB	
S	1032-3	EDB	
T	3031-1	EDB	
U	1040-2	EDB	
V	3004-1	EDB	
W	3013-1	EDB	
X	3018-1	EDB	
Z	3019-1	EDB	
a	3003-1	EDB	
b	4600-1	EDB	
c			
d	M3202	EDB	
e	4161-1	EDB	
f	4162-1	EDB	
g			
h	M3131	EDB	
j	M3135	EDB	
k	M3139	HSCB	
m			
n	M3144	HSCB	
p	M3145	HSCB	
r	M3142	HSCB	
s	M3143	HSCB	

CN2 (HL13)

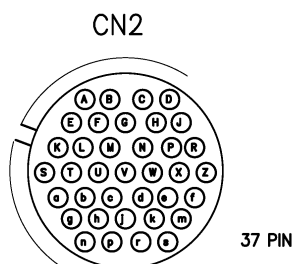
PIN NO	WIRE NO	TO	REMARK
A	3022-1	EDB	
B	M801-4	EDB	
C	3024-1	EDB	
D	M801-5	EDB	
E			
F	3026/1-3	EDB	GKW EMC 3x1.0
G	3026/2-3		
H	3026/3-3		
J	3026/S-3		
K			
L	4122	BECU T/B	
M	4126	BECU T/B	
N	4123-1	BECU T/B	
P			
R	4120	BECU T/B	
S	4226/1-2	BECU T/B	GKW EMC 2x1.0
T	4226/2-2		
U	4226/S-2		
V			
W	4228/1-2	BECU T/B	GKW EMC 2x1.0
X	4228/2-2		
Z	4228/S-2		
a			
b	4120A	BECU T/B	
c			
d	3591/1	BECU T/B	GKW EMC 2x1.0
e	3591/2		
f	3591/S		
g			
h	3590/1	BECU T/B	GKW EMC 2x1.0
j	3590/2		
k	3590/S		
m			
n			
p	4124	BECU T/B	
r	3592	BECU T/B	
s	4123-2	BECU T/B	

CN3 (HL14)

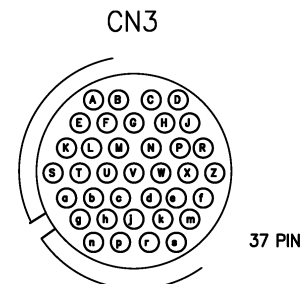
PIN NO	WIRE NO	TO	REMARK
A			
B			
C			
D			
E	M3027/1-1	TM CON#1	GKW EMC 4x1.0
F	M3027/2-1		
G	M3027/3-1		
H	M3027/4-1		
J	M3027/S-1		
K			
L	M3028/1-1	TM CON#2	GKW EMC 4x1.0
M	M3028/2-1		
N	M3028/3-1		
P	M3028/4-1		
R	M3028/S-1		
S			
T	M3029/1-1	TM CON#3	GKW EMC 4x1.0
U	M3029/2-1		
V	M3029/3-1		
W	M3029/4-1		
X	M3029/S-1		
Z			
a			
b	M3030/1-1	TM CON#4	GKW EMC 4x1.0
c	M3030/2-1		
d	M3030/3-1		
e	M3030/4-1		
f	M3030/S-1		
g			
h	3085/1-1	BECU T/B	GKW EMC 2x1.0
j	3085/2-1		
k	3085/S-1		
m			
n			
p			
r	M1201-2	EDB	
s	M1206-1	EDB	



CONNECTOR : VEG1D28-21SN



CONNECTOR : VEG1D28-21SW



CONNECTOR : VEG1D28-21SX

* NOTE

1. ALL CABLE EXCEPT SHIELD CABLE SHALL BE THE 3GKW-1.5
2. SPARE CABLE SHOULD BE TERMINATION APPLIED INSIDE FLEXIBLE TUBE

△						PRODUCT	KMRCL RS(3R) METRO CARS			
						REF DRG				
△						MATERIAL				
						HEAT TREAT.	APPD	K.C.SHASHIKANTH	27.01.2018	
△						SURFACE TREAT.	REVD	SADAHSIVAM.M	27.01.2018	
						TITLE	CHKD	V SARANYA	27.01.2018	
△						PIN ASSIGNMENT, VVVF INVERTER - M CAR	DRWN	SUSHILA PATIL	27.01.2018	
							SCALE	NTS	1 OF 2	0
1	DRAWING UPDATED & REVISED	DATE	13.02.18	BY	V.SARANYA	CHKD	G.VISHWANATH	APPD	K.C.SHASHIKANTH	
ALT.NO.	ECN NO/CHANGES	DATE		BY		DRG No.				ALT
						942-38706				1



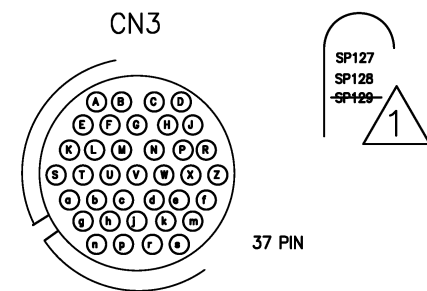
BEML LIMITED

942-38706

A3

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.	APPD	K.C.SHASHIKANTH		27.01.2018	
						SURFACE TREAT.	REVD	SADAHSIVAM.M		27.01.2018	
						TITLE	CHKD	V SARANYA		27.01.2018	
						PIN ASSIGNMENT, VVVF INVERTER - M CAR	DRWN	SUSHILA PATIL		27.01.2018	
	DRAWING UPDATED & REVISED	DATE 13.02.18	BY V/SARANYA	CHKD G.VISHWANATH	APPD K.C.SHASHIKANTH		SCALE			SHEET	Wt.(Kg)
							NTS			1 OF 2	0
ALT.NO.	ECN NO/CHANGES					BEML LIMITED NEW FRONTIERS. NEW DREAMS	DRG No.				ALT
							942-38706				

CN3 (HL14)			
PIN NO	WIRE NO	TO	REMARK
A			
B			
C			
D			
E	M3027/1-1	TM CON#1	GKW EMC 4x1.0
F	M3027/2-1		
G	M3027/3-1		
H	M3027/4-1		
J	M3027/S-1		
K			
L	M3028/1-1	TM CON#2	GKW EMC 4x1.0
M	M3028/2-1		
N	M3028/3-1		
P	M3028/4-1		
R	M3028/S-1		
S			
T	M3029/1-1	TM CON#3	GKW EMC 4x1.0
U	M3029/2-1		
V	M3029/3-1		
W	M3029/4-1		
X	M3029/S-1		
Z			
a			
b	M3030/1-1	TM CON#4	GKW EMC 4x1.0
c	M3030/2-1		
d	M3030/3-1		
e	M3030/4-1		
f	M3030/S-1		
g			
h	3085/1-1	BECU T/B	GKW EMC 2x1.0
j	3085/2-1		
k	3085/S-1		
m			
n			
p			
r	M1201-2	EDB	
s	M1206-1	EDB	



CONNECTOR : VEG1D28-21SX

1. ALL CABLE EXCEPT SHIELD CABLE SHALL BE THE 3GKW-1.5
2. SPARE CABLE SHOULD BE TERMINATION APPLIED INSIDE FLEXIBLE TUBE

[illegible]

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N1	0.025
N2	0.05
N3	0.1
N4	0.2
N5	0.4
N6	0.8
N7	1.6
N8	3.2
N9	6.3
N10	12.5
N11	25
N12	50
GRADE No.	
VALUE	
SYMBOL	

SURFACE
ROUGHNESS

5

4

3

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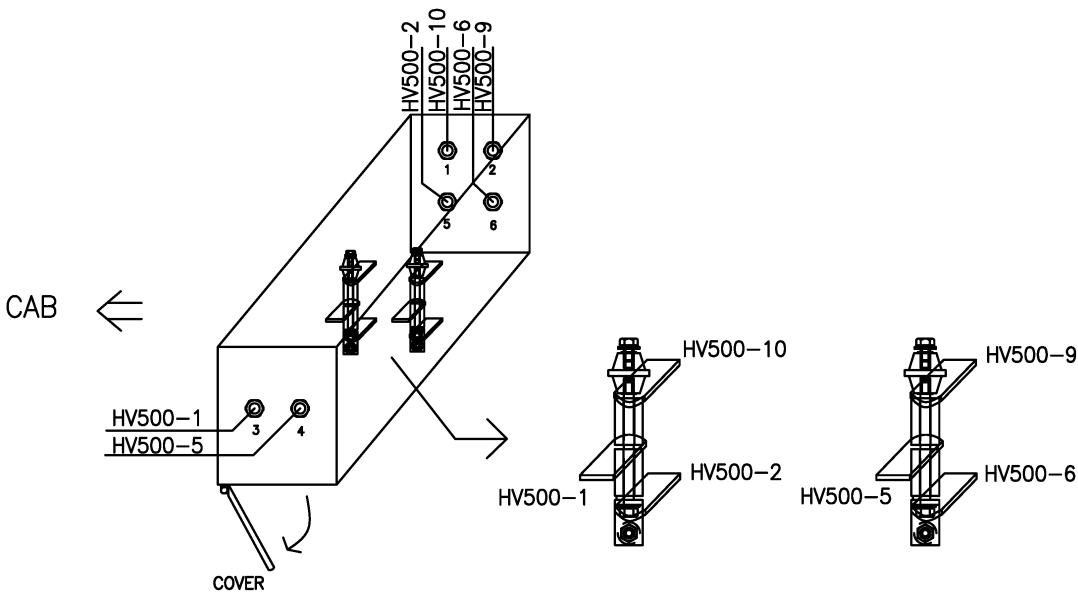
MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)								QUALITY OF WELD JOINTS REF, RD 230 MEDIUM		
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

COLLECTOR SHOE JUNCTION BOX 1

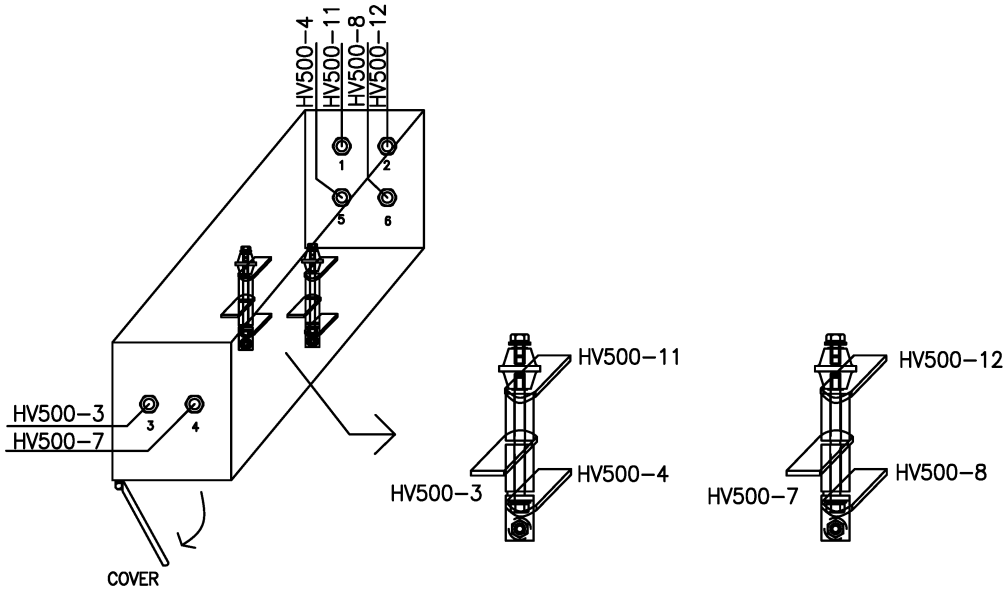
TERMINAL NO	WIRE NO	TO	REMARK
1	HV500-10	MS BOX A	
2	HV500-9	MS BOX A	
3	HV500-1	CCD FUSE BOX 1	
4	HV500-5	CCD FUSE BOX 1	
5	HV500-2	CCD FUSE BOX 2	
6	HV500-6	CCD FUSE BOX 2	

COLLECTOR SHOE JUNCTION BOX 2

TERMINAL NO	WIRE NO	TO	REMARK
1	HV500-11	MS BOX A	
2	HV500-12	MS BOX A	
3	HV500-3	CCD FUSE BOX 3	
4	HV500-7	CCD FUSE BOX 3	
5	HV500-4	CCD FUSE BOX 4	
6	HV500-8	CCD FUSE BOX 4	



CSJB1 (FRONT)



CSJB2 (REAR)

* NOTE

1. THE CABLE CONNECTED TO MS BOX SHALL BE 4GKW-AX-185
2. THE CABLE CONNECTED TO CCD FUSE BOX SHALL BE 4GKW-AX-J-185

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADAHSIVAM.M	27.01.2018	
						TITLE	PIN ASSIGNMENT, COLLECTOR SHOE JUNCTION BOX - M CAR		CHKD	V.SARANYA	27.01.2018
DRWN	SUSHILA PATIL	27.01.2018									
SCALE			SHEET	Wt.(Kg)							
NTS			1 OF 1	0							
						DRG No.				ALT	
						942-38707					
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEM LIMITED NEW FRONTIERS. NEW DREAMS					



BEML LIMITED

942-38707

A3

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SURFACE
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	ROUGHNESS
SYMBOL													

5

4

3

2

1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 - 6	6 - 30	30 - 120	120 - 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

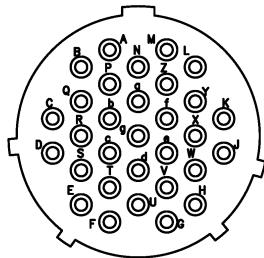
PRESSURE SWITCH BOX 1 (HL19)

PIN NO	WIRE NO	TO	BOX INSIDE	
			COMPONENT	PIN NO
A	M1231-1	BECU TB	MCUCMV	2
B	M2134-1	BEUC TB		1
C				
D				
E				
F				
G	M4181	TCMS TB	ASPS1(L11/1)	1
H	MB01-12	TCMS TB		3
J				
K	M4179	TCMS TB	BPPS1(B23)	1
L	MB02-4	TCMS TB		3
M				
N	M4179a	TCMS TB	BPPS2(B22)	1
P	MB02-5	TCMS TB		3
Q	4024	LTJB1	BPPS2(B22)	4
R	4026-1	BECU TB		6
S				
T	M4135a	TCMS TB	PBPS1(B15/1)	1
U	MB01	TCMS TB		3
V	4154	EDB		4
W	4155b	LTJB1		6
X				
Y				1
Z				3
a				4
b				6
c				
d	M4180	TCMS TB	BCPS1(B07/1)	1
e	MB01-10	TCMS TB		3
f	4161-2	EDB		4
g	4360-1	EDB		6

PRESSURE SWITCH BOX 2 (HL5)

PIN NO	WIRE NO	TO	BOX INSIDE	
			COMPONENT	PIN NO
A	M4182	TCMS TB	ASPS2(L11/2)	1
B	MB01-13	TCMS TB		3
C				
D	M4155a	TCMS TB	PBPS2(B15/2)	1
E	MB01-1	TCMS TB		3
F	4155	LTJB2		4
G	4154-1	EDB		6
H				
J	M4183	TCMS TB	BCPS2(B07/2)	1
K	MB01-11	TCMS TB		3
L	4162-2	EDB		4
M	4360A-1	EDB		6
N				
P				
Q				
R				
S				
T				
U				
V				
W				
X				
Y				
Z				
a				
b				
c				
d				
e				
f				
g				

SP106-1
SP109
SP119
SP143
SP145
SP146



REAR VIEW OF PIN INSERT
CONNECTOR : 31PIN

* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 3GKW-1.5

SP4026
SP108
SP126
SP144
SP4179
SP4181
SP147
SP148

						PRODUCT	KMRCL RS(3R) METRO CARS							
						REF DRG								
						MATERIAL								
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	29.01.2018				
						SURFACE TREAT.		REVD	SADAHSIVAM.M	29.01.2018				
						TITLE		CHKD	V.SARANYA	29.01.2018				
						PIN ASSIGNMENT, PRESSURE SWITCH BOX- M CAR	DRWN	SUSHILA PATIL	29.01.2018					
	DRAWING UPDATED	DATE	10.03.2018	BY	SUSHILA PATIL		CHKD	SARANYA	APPD	KCS	SCALE		SHEET	Wt.(Kg)
											NTS		1 OF 1	0
ALT.NO.	ECN NO/CHANGES							DRG No.			ALT			
							BEML LIMITED			942-38709				

5

4

3

2

1

A3

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SURFACE ROUGHNESS													

D

C

B

A

5

4

3

2

1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

BCU1

BCU2

PIN NO	WIRE NO	TO	REMARK
A1	M4220/1-1	BECU T/B	GKW EMC 3x1.0
A2	M4220/3-1	BECU T/B	GKW EMC 3x1.0
A3			
A4	M4141-1	EDB	
A5	M4220/2-1	BECU T/B	GKW EMC 3x1.0
A6			
A7			
A8	M4142-1	EDB	
A9			
A10			
A11			
A12			
B1	M4223b/1	BECU T/B	GKW EMC 2x1.0
B4	M4223a/2	BECU T/B	
B7	M4223b/2	BECU T/B	
B8	M4223a/1	BECU T/B	
C1	M4221/2	BECU T/B	
C2	M4222/2	BECU T/B	GKW EMC 2x1.0
C3	M4223/2	BECU T/B	
C4			
C5	M4221/1	BECU T/B	
C6	M4222/1	BECU T/B	
C7	M4223/1	BECU T/B	
C8			
C9			
C10			
C11			
C12			

PIN NO	WIRE NO	TO	REMARK
A1	M4354/1	BECU T/B	GKW EMC 3x1.0
A2	M4354/3	BECU T/B	GKW EMC 3x1.0
A3			
A4	M4141-2	EDB	
A5	M4354/2	BECU T/B	GKW EMC 3x1.0
A6		BECU T/B	GKW EMC 2x1.0
A7			
A8	M4142-2	EDB	
A9			
A10			
A11			
A12			
B1	M4357b/1	BECU T/B	GKW EMC 2x1.0
B4	M4357a/2	BECU T/B	
B7	M4357b/2	BECU T/B	
B8	M4357a/1	BECU T/B	
C1	M4356/2	BECU T/B	
C2	M4355/2	BECU T/B	GKW EMC 2x1.0
C3	M4357/2	BECU T/B	
C4			
C5	M4356/1	BECU T/B	
C6	M4355/1	BECU T/B	
C7	M4357/1	BECU T/B	
C8			
C9			
C10			
C11			
C12			

SP131

- 1) M4354/S,M4355/S,M4356/S,M4357/S
M4357a/S,M4357b/S
SHALL BE TEMINATION APPLIED.
- 2) PINS A5 AND A6 TO BE SHORTED

SP130

- 1) M4220/S,M4222/S,M4221/S,M4223/S,M4223a/S,M4223b/S
SHALL BE TEMINATION APPLIED.
- 2) PINS A5 AND A6 TO BE SHORTED

* NOTE

1. ALL CABLE EXCEPT SHIELD CABLE SHALL BE THE 3GKW-1.5

△						PRODUCT	KMRCL RS(3R) METRO CARS			
△						REF DRG				
△						MATERIAL				
△						HEAT TREAT.	APPD	K.C.SHASHIKANTH	09.11.2017	
△						SURFACE TREAT.	REVD	SADAHSIVAM.M	09.11.2017	
△						TITLE	CHKD	V.SARANYA	09.11.2017	
△						PIN ASSIGNMENT, BCU - M CAR	DRWN	SUSHILA PATIL	09.11.2017	
△							SCALE	NTS	1 OF 1	0
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	DRG No.				ALT
						BEML LIMITED				0
						942-38710				

D

C

B

A

A3

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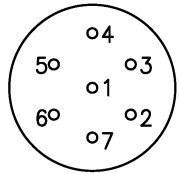
GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SURFACE ROUGHNESS													

ASCO 1 (HL11)

PIN NO	WIRE NO	TO	REMARK
4	M4176	TCMS T/B	
5	MB01-8	TCMS T/B	

ASCO 2 (HL18)

PIN NO	WIRE NO	TO	REMARK
4	M4185	TCMS T/B	
5	MB01-9	TCMS T/B	



CONNECTOR : N63583

* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 3GKW-1.5
2. ASCO : BOGIE AIR SUSPENSION ISOLATION
3. EPIC : COMPLETE BRAKE ISOLATION
4. SR : SUPPLY RESVIOR

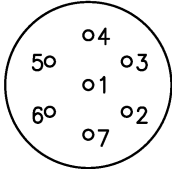
MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315–1000	1000–2000	2000–4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

EPIC1 (HL9)

PIN NO	WIRE NO	TO	REMARK
4	M4177	TCMS T/B	
5	MB01-2	TCMS T/B	

EPIC2 (HL17)

PIN NO	WIRE NO	TO	REMARK
4	M4361	TCMS T/B	
5	MB01-3	TCMS T/B	



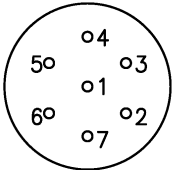
CONNECTOR : N63583

SR1 (HL10)

PIN NO	WIRE NO	TO	REMARK
4	M4184	TCMS T/B	
5	MB01-6	TCMS T/B	

SR2 (HL17)

PIN NO	WIRE NO	TO	REMARK
4	M4184a	TCMS T/B	
5	MB01-7	TCMS T/B	



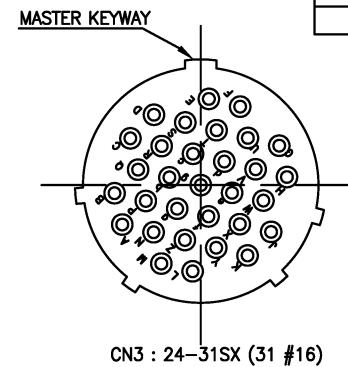
CONNECTOR : N63583

△						PRODUCT	KMRCL RS(3R) METRO CARS			
△						REF DRG				
△						MATERIAL				
△						HEAT TREAT.		APPD	K.C.SHASHIKANTH	29.01.2018
△						SURFACE TREAT.		REVD	SADAHSIVAM.M	29.01.2018
△						TITLE PIN ASSIGNMENT, ASCO, EPIC, SR - M CAR	CHKD	V.SARANYA	29.01.2018	
△							DRWN	SUSHILA PATIL	29.01.2018	
△							SCALE	NTS	SHEET	Wt.(Kg)
△							NTS		1 OF 1	0
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	DRG No.				ALT
						BEML LIMITED				0

942-38711

						PRODUCT	KMRL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	29.01.2018	
						SURFACE TREAT.		REVD	SADHASIVAM.M	29.01.2018	
						TITLE	CHKD		V.SARANYA	29.01.2018	
						PIN ASSIGNMENT, LTJB1 - M CAR	DRWN		SUSHILA PATIL	29.01.2018	
	X5 & X10 Pin details updated	12.04.2018	AMBIKA	GIRISH	KCS		SCALE		SHEET	Wt.(Kg)	
							NTS		1 OF 3	0	
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEM LIMITED NEW FRONTIERS. NEW DREAMS	DRG No.			ALT	
							942-38712				

CN3			
PIN NO	WIRE NO	TO	REMARK
A	SP001	EDB	
B	SP002	EDB	
C	SP003	EDB	
D	SP004	EDB	
E	SP005	EDB	
F	SP006	EDB	
G	SP007	EDB	
H	SP008	BECU T/B	
J	SP009	BECU T/B	
K	SP010	BECU T/B	
L	Q 3085/1	BECU T/B	GKW EMC 2x1.0
M	Q 3085/2		
N	Q 3085/S		
P	Q SP026/1	BECU T/B	GKW EMC 2x1.0
Q	Q SP026/2		
R	Q SP027/1	BECU T/B	GKW EMC 2x1.0
S	Q SP027/2		
T	Q SP028/1	BECU T/B	GKW EMC 2x1.0
U	Q SP028/2		
V	Q SP029/1	BECU T/B	GKW EMC 2x1.0
W	Q SP029/2		
X	Q SP030/1	BECU T/B	GKW EMC 2x1.0
Y	Q SP030/2		
Z	Q SP031/1	BECU T/B	GKW EMC 2x1.0
a	Q SP031/2		
b	Q 9078/1	SPK(R2)	BS FR EMC 2X1.5
c	Q 9078/2		
d	Q 9076/1	AAU	BS FR EMC 2X1.5
e	Q 9076/2		
f	J		
g			



SP034/1,2,S
SP035/1,2,S
SPCAN1/1,2,3,S
SPCAN2/1,2,3,S

* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 3GKW-1.5

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GRADE No.	N1	0.025
VALUE	N2	0.05
SYMBOL	N3	0.1
	N4	0.2
	N5	0.4
	N6	0.8
	N7	1.6
	N8	3.2
	N9	6.3
	N10	12.5
	N11	25
	N12	50
SURFACE		
ROUGHNESS		

5

4

3

2

1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 - 6	6 - 30	30 - 120	120 - 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

CN4

PIN NO	WIRE NO	TO	REMARK
A	SP032/1	BECU T/B	GKW EMC 2x1.0
B	SP032/2		
C	SP033/1		
D	SP033/2	EDB	GKW EMC 2x1.0
E	9240/1	LTJB2	GKW EMC 2x1.0
F	9240/2		
G	9241/1	LTJB2	GKW EMC 2x1.0
H	9241/2		
J	9242/1	LTJB2	GKW EMC 2x1.0
K	9242/2		
L	9243/1	LTJB2	GKW EMC 2x1.0
M	9243/2		
N	9245/1	LTJB2	GKW EMC 2x1.0
P	9245/2		
Q	9197	LTJB2	100F1111-1.34
R	9198		
S	9199		
T	9200		
U	9217		
V	9218	LTJB2	GKW EMC 2x1.0
W	9219		
X	9220	LTJB2	GKW EMC 2x1.0
Y	SP020/1		
Z	SP020/2	EDB	GKW EMC 2x1.0
a	SP020/S		
b	SP021/1		
c	SP021/2	EDB	GKW EMC 2x1.0
d	SP021/S		
e	SP022/1		
f	SP022/2	EDB	GKW EMC 2x1.0
g	SP022/S		

CN5

PIN NO	WIRE NO	TO	REMARK
A	SP023/1	EDB	GKW EMC 2x1.0
B	SP023/2		
C	SP023/S		
D	SP024/1	EDB	GKW EMC 2x1.0
E	SP024/2		
F	SP025/1	EDB	GKW EMC 2x1.0
G	SP025/2		
H			
J			
K			
L			
M			
N			
P			
Q			
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
f			
g			

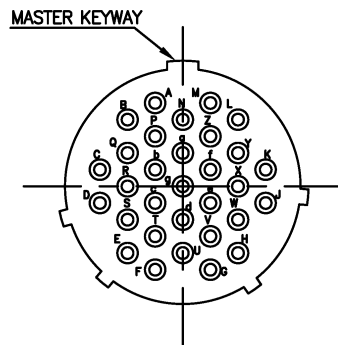
CN6

PIN NO	WIRE NO	TO	REMARK
A	SP011	BECU TB	
B	SP012	BECU TB	
C	SP013	BECU TB	
D	SP014	BECU TB	
E	SP015	LTJB2	
F			
G			
H			
J			
K			
L			
M			
N			
P			
Q			
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
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g			

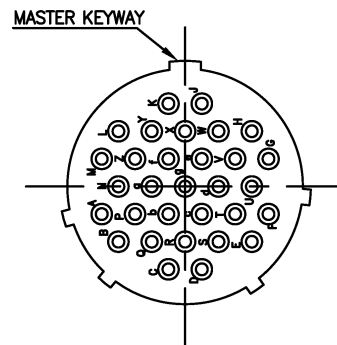
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SP024/S,SP025/S SHALL BE CONNECTED TO 'H' PIN AFTER SLEEVE JOINT.

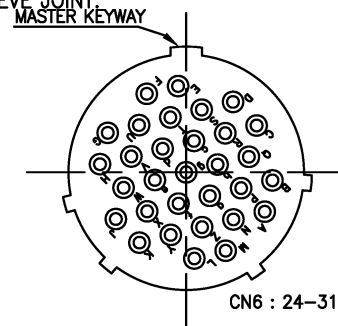
9244/1,2,S
9246/1,2,S



CN4 : 24-31P (31 #16)



CN5 : 24-31PW (31 #16)



CN6 : 24-31PX (31 #16)

SP116
SP117
SP118
BECU T/B

* NOTE
1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 3GKW-1.5

△						PRODUCT	KMRCL RS(3R) METRO CARS			
						REF DRG				
△						MATERIAL				
						HEAT TREAT.	APPD	K.C.SHASHIKANTH	29.01.2018	
△						SURFACE TREAT.	REVD	SADHASIVAM.M	29.01.2018	
						TITLE	CHKD	V.SARANYA	29.01.2018	
△						PIN ASSIGNMENT, LTJB1 - M CAR	DRWN	SUSHILA PATIL	29.01.2018	
							SCALE			
1	X5 & X10 Pin details updated	12.04.2018	BY AMBIKA	CHKD GIRISH	APPD KCS	NTS				
ALT.No.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	DRG No.	SHEET 2 OF 3		Wt.(Kg)	0
b e m l BEML LIMITED							942-38712		ALT	0

5

4

3

2

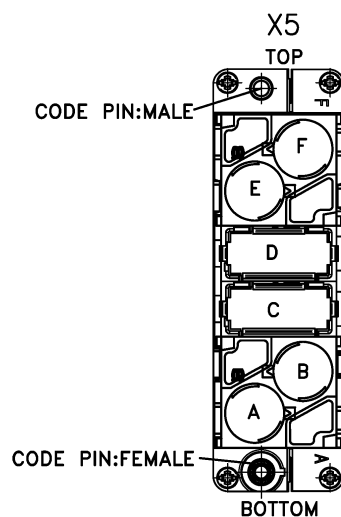
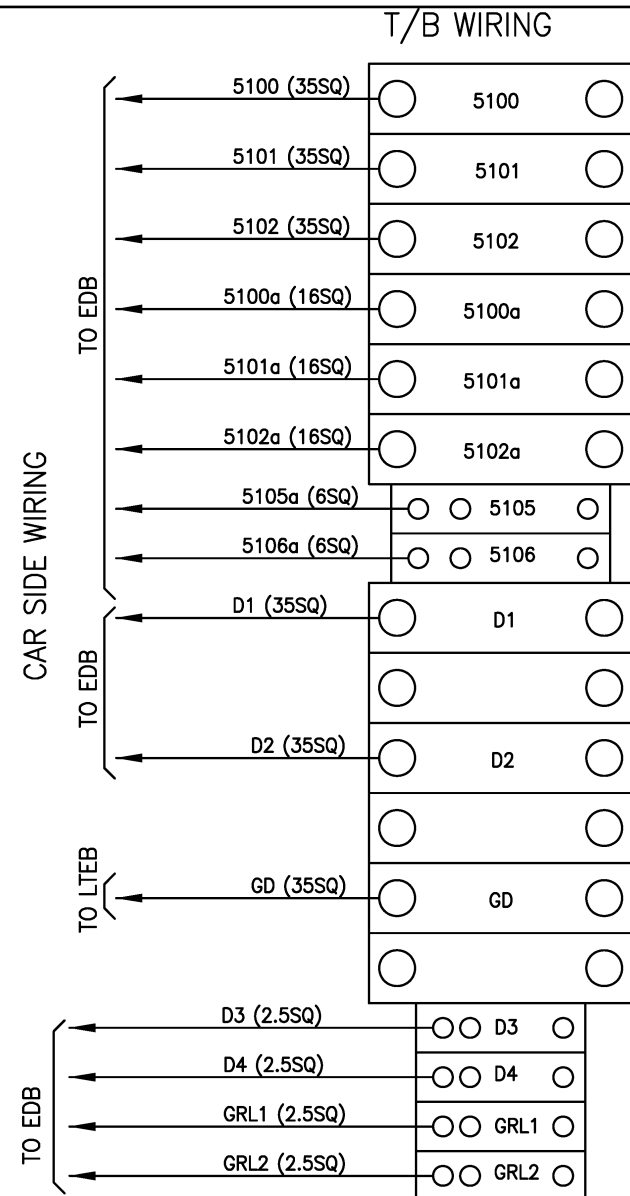
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A3

GRADE No.	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1
VALUE	50	25	12.5	6.3	3.2	1.6	0.8	0.4	0.2	0.1	0.05	0.025
SYMBOL												

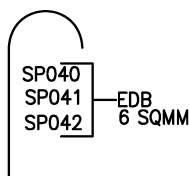
MARK										
38 RAILCAT										
38 RAILCAT										
38 RAILCAT										
38 RAILCAT										
38 RAILCAT										
38 RAILCAT										

PIN NO	WIRE NO	TO	REMARK
f1	9222/1	L3 SWITCH_2	12584038 (RADOX RAILCAT CAT5e)
f2	9222/2		
f3	9222/3		
f4	9222/4		
e1	9224/1	L3 SWITCH_2	12584038 (RADOX RAILCAT CAT5e)
e2	9224/2		
e3	9224/3		
e4	9224/4		
d1	9230/1	L3 SWITCH_2	12584038 (RADOX RAILCAT CAT5e)
d2	9230/2		
d3	9230/3		
d4	9230/4		
c1	9228/1	L3 SWITCH_1	12584038 (RADOX RAILCAT CAT5e)
c2	9228/2		
c3	9228/3		
c4	9228/4		
b1	9227/1	L3 SWITCH_1	12584038 (RADOX RAILCAT CAT5e)
b2	9227/2		
b3	9227/3		
b4	9227/4		
a1	9226/1	L3 SWITCH_1	12584038 (RADOX RAILCAT CAT5e)
a2	9226/2		
a3	9226/3		
a4	9226/4		



Module a, b

1-TX+ (YELLOW)
2-RX+ (WHITE)
3-TX- (ORANGE)
4-RX- (BLUE)



					PRODUCT	KMRCL RS(3R) METRO CARS					
					REF DRG						
					MATERIAL						
					HEAT TREAT.	APPD	K.C.SHASHIKANTH			29.01.2017	
					SURFACE TREAT.	REVD	SADHASIVAM.M			29.01.2017	
					TITLE	CHKD	V.SARANYA			29.01.2017	
					PIN ASSIGNMENT, LTJB1 - M CAR	DRWN	SUSHILA PATIL			29.01.2017	
	X5 & X10 Pin details updated	12.04.2018	AMBIKA	GRISH		KCS	SCALE			SHEET	Wt.(Kg)
							NTS			3 OF 3	0
ALT.NO.	ECN NO/CHANGES					DRG No.					ALT
					BEML LIMITED NEW FRONTIERS. NEW DREAMS	942-38712					

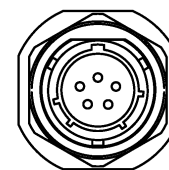
PIN NO	WIRE NO	TO	REMARK	
J	M4216/1-1	BECU T/B		
C	M4216/2-1			
B	M4216a/1-1			
H	M4216a/2-1			
J	M4216/1	SPEED SENSOR 2	RED	MAKER SUPPLY (KNORR)
C	M4216/2		WHITE	
			SHIELD	
B	M4216a/1		ORANGE	
H	M4216a/2		GREEN	

PEED SENSOR CONNECTOR 4

PIN NO	WIRE NO	TO	REMARK	
J	M4218/1-1	BECU T/B		
C	M4218/2-1			
B	M4218a/1-1			
H	M4218a/2-1			
J	M4218/1	SPEED SENSOR 4	RED	MAKER SUPPLY (KNORR)
C	M4218/2		WHITE	
			SHIELD	
B	M4218a/1		ORANGE	
H	M4218a/2		GREEN	

PIN NO	WIRE NO	TO	REMARK	
J	M4217/1-1	BECU T/B		
C	M4217/2-1			
B	M4217a/1-1			
H	M4217a/2-1			
J	M4217/1	SPEED SENSOR 3	RED	MAKER SUPPLY (KNORR)
C	M4217/2		WHITE	
			SHIELD	
B	M4217a/1		ORANGE	
H	M4217a/2		GREEN	

PIN NO	WIRE NO	TO	REMARK	
J	M4218/1-1	BECU T/B		
C	M4218/2-1			
B	M4218a/1-1			
H	M4218a/2-1			
J	M4218/1	SPEED SENSOR 4	RED	MAKER SUPPLY (KNORR)
C	M4218/2		WHITE	
			SHIELD	
B	M4218a/1		ORANGE	
H	M4218a/2		GREEN	



					PRODUCT	KMRL RS(3R) METRO CARS					
					REF DRG						
					MATERIAL						
					HEAT TREAT.	APPD	K.C.SHASHIKANTH		30.01.2018		
					SURFACE TREAT.	REVD	SADAHSIVAM.M		30.01.2018		
					TITLE	CHKD	V.SARANYA		30.01.2018		
	DRAWING UPDATED	05.02.2018	BY SUSHILA	CHKD SARNAYA.V	K.C.S	PIN ASSIGNMENT - SPEED SENSOR CONNECTOR (MC)		DRWN	SUSHILA PATIL		30.01.2018
						SCALE			SHEET	Wt.(Kg)	
						NTS			1 OF 1	0	
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	 beml BEML LIMITED NEW FRONTIERS. NEW DREAMS		DRG No.		ALT	
								942-38714			

* NOTE

1. BECU T/B SIDE CABLE SHALL BE THE GKW EMC 2 x1.0
2. SPEED SENSOR SIDE CABLE SUPPLIED BY BRAKE SUPPLIER

GRADE No.	N12	N11	N10	N9	N8	N7	N6	N5	N4	N3	N2	N1
VALUE	50	25	12.5	6.3	3.2	1.6	0.8	0.4	0.2	0.1	0.05	0.025
SYMBOL												

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADAHSIVAM.M	27.01.2018	
						TITLE			CHKD	V.SARANYA	27.01.2018
						PIN ASSIGNMENT, BRAKE RESISTOR, FILTER REACTOR - M CAR	DRWN		SUSHILA PATIL	27.01.2018	
							SCALE		SHEET	Wt.(Kg)	
							NTS		1 OF 1	0	
						DRG No.					
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPO	 BEML LIMITED NEW FRONTIERS. NEW DREAMS	942-38715			ALT	

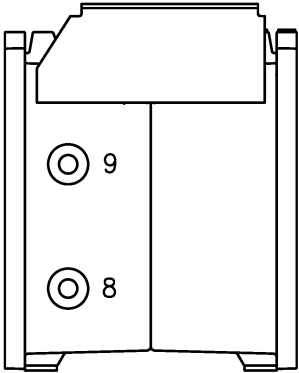
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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
ROUGHNESS													

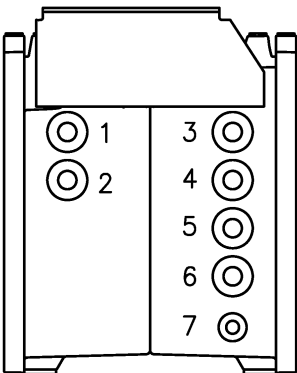
* NOTE
1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 4GKW-AX-185

MAIN SWITCH BOX A

TERMINAL NO	WIRE NO	TO	REMARK
1	HV500a-2	STINGER BOX (L)	
2	HV500a-1	STINGER BOX (R)	
3	HV500-10	CSJB 1	
4	HV500-9	CSJB 1	
5	HV500-11	CSJB 2	
6	HV500-12	CSJB 2	
7	HV500-13	SURGE ARRESTER	4GKW-AX-35
8	HV501-2	MS BOX B	
9	HV501-3	MS BOX B	



<LEFT SIDE>



<RIGHT SIDE>

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADAHSIVAM.M	27.01.2018	
						TITLE	PIN ASSIGNMENT, MAIN SWITCH BOX - M CAR	CHKD	V.SARANYA	27.01.2018	
								DRWN	SUSHILA PATIL	27.01.2018	
								SCALE		SHEET	Wt.(Kg)
								NTS		1 OF 2	0
								DRG No.	942-38716		
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEML LIMITED NEW FRONTIERS. NEW DREAMS					

 **BEML LIMITED**
NEW FRONTIERS. NEW DREAMS

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 - 6	6 - 30	30 - 120	120 - 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
ROUGHNESS													

D

C

B

A

5

4

3

2

1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

D

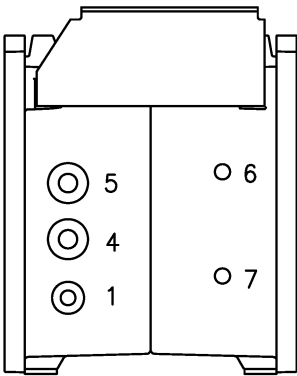
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B

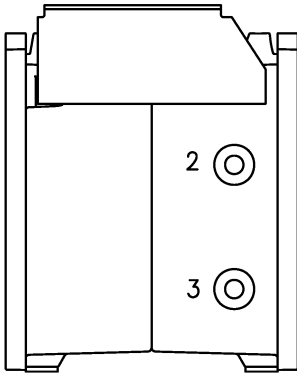
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MAIN SWITCH BOX B

TERMINAL NO	WIRE NO	TO	REMARK
1	HV501-1	HTJB	4GKW-AX-70
2	HV501-2	MS BOX A	4GKW-AX-185
3	HV501-3	MS BOX A	4GKW-AX-185
4	HV502-1	HSCB	4GKW-AX-185
5	HV502-2	HSCB	4GKW-AX-185
6	HV799-3	VVF INVERTER	4GKW-AX-6
	HV533	VVF INVERTER	4GKW-AX-6
7	HV799-4	VVF INVERTER	4GKW-AX-6
	HV534	VVF INVERTER	4GKW-AX-6



<LEFT SIDE>



<RIGHT SIDE>

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADAHSIVAM.M	27.01.2018	
						TITLE	PIN ASSIGNMENT, MAIN SWITCH BOX - M CAR	CHKD	V.SARANYA	27.01.2018	
								DRWN	SUSHILA PATIL	27.01.2018	
								SCALE		SHEET	Wt.(Kg)
								NTS		2 OF 2	0
								DRG No.			
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEM LIMITED					
						942-38716					0

5

4

3

2

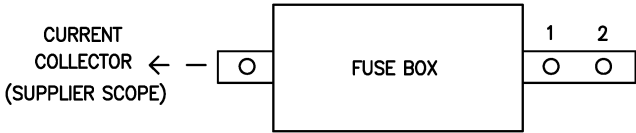
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A3

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SYMBOL													

SURFACE
ROUGHNESS



TERMINAL ARRANGEMEENT

* NOTE
1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 4GKW--AX--J--185

CURRENT COLLECTOR FUSE BOX 1

TERMINAL NO	WIRE NO	TO	REMARK
1	HV500--1	CSJB 1	
2	HV500--5	CSJB 1	

CURRENT COLLECTOR FUSE BOX 2

TERMINAL NO	WIRE NO	TO	
1	HV500--2	CSJB 1	
2	HV500--6	CSJB 1	

CURRENT COLLECTOR FUSE BOX 3

TERMINAL NO	WIRE NO	TO	REMARK
1	HV500--3	CSJB 2	
2	HV500--7	CSJB 2	

CURRENT COLLECTOR FUSE BOX 4

TERMINAL NO	WIRE NO	TO	
1	HV500--4	CSJB 2	
2	HV500--8	CSJB 2	

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

						PRODUCT	KMRCL RS(3R) METRO CARS			
						REF DRG				
						MATERIAL				
						HEAT TREAT.	APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.	REVD	SADAHSIVAM.M	27.01.2018	
						TITLE PIN ASSIGNMENT, CURRENT COLLECTOR FUSE BOX - M CAR	CHKD	V.SARANYA	27.01.2018	
							DRWN	SUSHILA PATIL	27.01.2018	
							SCALE		SHEET	Wt.(Kg)
						NTS			1 OF 1	0
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	DRG No.			ALT	
						BEML LIMITED			942-38717	

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50
SYMBOL												

SURFACE
ROUGHNESS

5

4

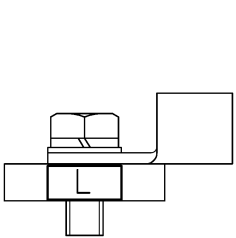
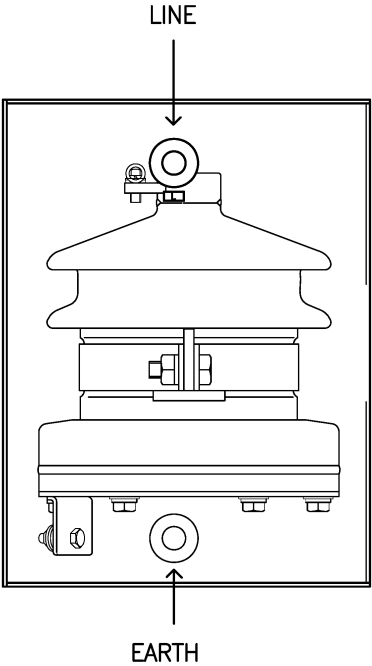
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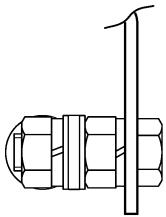
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MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)								QUALITY OF WELD JOINTS REF, RD 230 MEDIUM		
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

TERMINAL NO	WIRE NO	TO	REMARK
LINE	HV500-13	MS BOX A	
EARTH	PE500	CAR BODY	



LINE TERMINAL



EARTH TERMINAL

* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 4GKW-AX-35

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADAHSIVAM.M	27.01.2018	
						TITLE	CHKD	V.SARANYA	27.01.2018		
						PIN ASSIGNMENT, SURGE ARRESTOR - M CAR	DRWN	SUSHILA PATIL	27.01.2018		
							SCALE			SHEET	Wt.(Kg)
							NTS		1 OF 1	0	
							DRG No.				
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEM LIMITED NEW FRONTIERS. NEW DREAMS	942-38718				

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4

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1

A3

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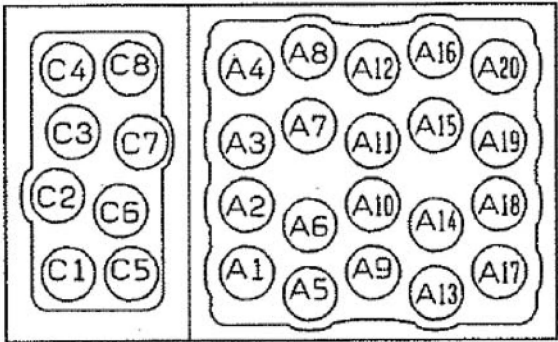
GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	VALUE	SYMBOL
	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50		
SURFACE ROUGHNESS														

SP152
SP132
SP133
SP134
SP3131
SP3135
1053

- * NOTE
1. ALL CABLE FOR CONNECTOR SHALL BE THE 3GKW-1.5
 2. ALL CABLE FOR TERMINAL SHALL BE THE 4GKW-AX-185
 3. M3144 & M3145 CABLE SHALL BE 3GKW-2.5

HIGH SPEED CIRCUIT BREAKER (HL15)

PIN NO	EXTERNAL LINE NO.	DESTINATION
A1	M3020-3	EDB
A2		
A3	M3135A	EDB
A4		
A5	M3020-4	EDB
A6		
A7	M3139	VWF INVERTER
A8		
A9	M1201-3	EDB
A10		
A11	M1207a	EDB
A12		
A13	M3142	VWF INVERTER
A14	M1201-4	EDB
A15	M3143	VWF INVERTER
A16	1208	EDB
A17		
A18		
A19		
A20		
C1	M3144	VWF INVERTER (3GKW-2.5)
C2	M3145	
C3		
C4		
C5	M3020-5	EDB
C6		
C7	M3131A	EDB
C8		

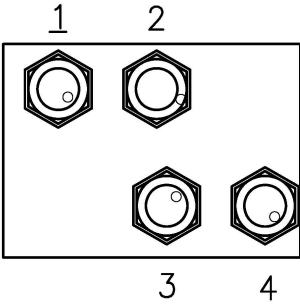


CONNECTOR PLUG : 9370100427

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

TERMINAL

TERMINAL NO	WIRE NO	TO	REMARK
1	HV502-1	MS BOX B	
2	HV502-2	MS BOX B	
3	HV510-1	VWF INVERTER	
4	HV510-2	VWF INVERTER	



TERMINAL ARRANGEMENT

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADAHSIVAM.M	27.01.2018	
						TITLE	PIN ASSIGNMENT, HSCB - M CAR		CHKD	V.SARANYA	27.01.2018
DRWN	SUSHILA PATIL	27.01.2018									
SCALE		SHEET	Wt.(Kg)								
NTS		1 OF 1	0								
						DRG No.				ALT	
						942-38719					
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	 BEML LIMITED NEW FRONTIERS. NEW DREAMS					

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N1	0.025	
N2	0.05	
N3	0.1	
N4	0.2	
N5	0.4	
N6	0.8	
N7	1.6	
N8	3.2	
N9	6.3	
N10	12.5	
N11	25	
N12	50	
GRADE No.		
VALUE		
SYMBOL		

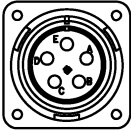
SURFACE
ROUGHNESS

T/M CONNECTOR 1

PIN NO	WIRE NO	TO	REMARK
A	M3027/1-1	VWF INVERTER	
B	M3027/2-1		
C	M3027/3-1		
D	M3027/4-1		
E	M3027/S-1		
A	M3027/1-2	T/M SPEED SENSOR 1	
B	M3027/2-2		
C	M3027/3-2		
D	M3027/4-2		
E	M3027/S-2		

T/M CONNECTOR 3

PIN NO	WIRE NO	TO	REMARK
A	M3029/1-1	VWF INVERTER	
B	M3029/2-1		
C	M3029/3-1		
D	M3029/4-1		
E	M3029/S-1		
A	M3029/1-2	T/M SPEED SENSOR 3	
B	M3029/2-2		
C	M3029/3-2		
D	M3029/4-2		
E	M3029/S-2		



VEHICLE SIDE (RECEPTACLE)
FRITS4100A18-11PB1N5F6N380



T/M SIDE (PLUG)
FRITS4106A18-11SB1N2N5F6

- * NOTE
1. VWF INVERTER SIDE CABLE SHALL BE THE GKW EMC 4x1.0
 2. T/M SPEED SENSOR SIDE CABLE SUPPLIED BY MELCO

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

T/M CONNECTOR 2

PIN NO	WIRE NO	TO	REMARK
A	M3028/1-1	VWF INVERTER	
B	M3028/2-1		
C	M3028/3-1		
D	M3028/4-1		
E	M3028/S-1		
A	M3028/1-2	T/M SPEED SENSOR 2	
B	M3028/2-2		
C	M3028/3-2		
D	M3028/4-2		
E	M3028/S-2		

T/M CONNECTOR 4

PIN NO	WIRE NO	TO	REMARK
A	M3030/1-1	VWF INVERTER	
B	M3030/2-1		
C	M3030/3-1		
D	M3030/4-1		
E	M3030/S-1		
A	M3030/1-2	T/M SPEED SENSOR 4	
B	M3030/2-2		
C	M3030/3-2		
D	M3030/4-2		
E	M3030/S-2		

△						PRODUCT	KMRCL RS(3R) METRO CARS			
						REF DRG				
△						MATERIAL				
						HEAT TREAT.	APPD	KC SHAHSIKANTH	30.01.2018	
△						SURFACE TREAT.	REVD	SADHASIVAM M	30.01.2018	
						TITLE	CHKD	V.SARANYA	30.01.2018	
△						PIN ASSIGNMENT, T/M CONNECTOR - M CAR	DRWN	SUSHILA PATIL	30.01.2018	
							SCALE		SHEET	Wt.(Kg)
△							NTS		1 OF 2	0
							DRG No.			ALT
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEML LIMITED	942-38720			

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N1	0.025	
N2	0.05	
N3	0.1	
N4	0.2	
N5	0.4	
N6	0.8	
N7	1.6	
N8	3.2	
N9	6.3	
N10	12.5	
N11	25	
N12	50	
GRADE No.		
VALUE		
SYMBOL		

SURFACE
ROUGHNESS

5

4

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MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)								QUALITY OF WELD JOINTS REF, RD 230 MEDIUM		
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

T/M CONNECTOR 1

TERMINAL NO	WIRE NO	TO	REMARK
1	EM500E1-2	VVF INVERTER	EARTH
2	EM553-2	VVF INVERTER	W
3	EM551-2	VVF INVERTER	V
4	EM550-2	VVF INVERTER	U
5	EM500E1-1	T/M 1	EARTH
6	EM553-1	T/M 1	W
7	EM551-1	T/M 1	V
8	EM550-1	T/M 1	U

T/M CONNECTOR 2

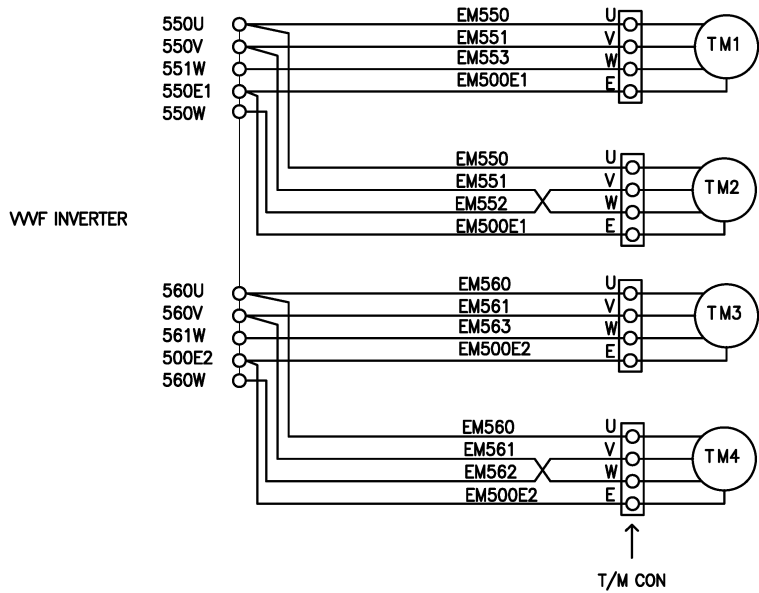
TERMINAL NO	WIRE NO	TO	REMARK
1	EM500E1-4	VVF INVERTER	EARTH
2	EM551-4	VVF INVERTER	V
3	EM552-2	VVF INVERTER	W
4	EM550-4	VVF INVERTER	U
5	EM500E1-3	T/M 2	EARTH
6	EM551-3	T/M 2	W
7	EM552-1	T/M 2	V
8	EM550-3	T/M 2	U

T/M CONNECTOR 3

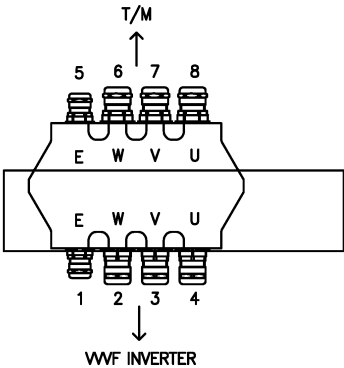
TERMINAL NO	WIRE NO	TO	REMARK
1	EM500E2-2	VVF INVERTER	EARTH
2	EM563-2	VVF INVERTER	W
3	EM561-2	VVF INVERTER	V
4	EM560-2	VVF INVERTER	U
5	EM500E2-1	T/M 3	EARTH
6	EM563-1	T/M 3	W
7	EM561-1	T/M 3	V
8	EM560-1	T/M 3	U

T/M CONNECTOR 4

TERMINAL NO	WIRE NO	TO	REMARK
1	EM500E2-4	VVF INVERTER	EARTH
2	EM561-4	VVF INVERTER	V
3	EM562-2	VVF INVERTER	W
4	EM560-4	VVF INVERTER	U
5	EM500E2-3	T/M 4	EARTH
6	EM561-3	T/M 4	W
7	EM562-1	T/M 4	V
8	EM560-3	T/M 4	U



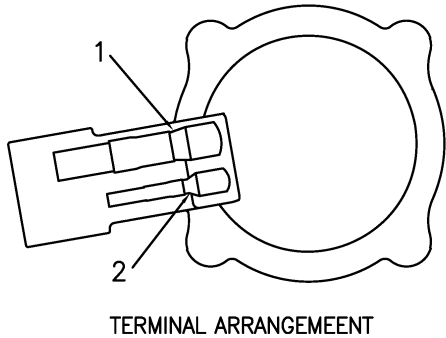
* NOTE
1. VVF INVERTER SIDE CABLE
- U,V,W CABLE SHALL BE 4GKW-AX-95



						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	KC SHAHSIKANTH	30.01.2018	
						SURFACE TREAT.		REVD	SADHASIVAM M	30.01.2018	
						TITLE		CHKD	V.SARANYA	30.01.2018	
						PIN ASSIGNMENT, T/M CONNECTOR - M CAR		DRWN	SUSHILA PATIL	30.01.2018	
							SCALE			SHEET	Wt.(Kg)
							NTS			2 OF 2	0
							DRG No.				
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEML LIMITED NEW FRONTIERS. NEW DREAMS	942-38720				

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SURFACE ROUGHNESS													



TERMINAL ARRANGEMEENT

EARTH BRUSH 1

TERMINAL NO	WIRE NO	TO	REMARK
1	MG301	HTEB	4GKW-AX-J-185
2	MBG350	BOGIE FRAME 1	MAKER SUPPLY

EARTH BRUSH 2

TERMINAL NO	WIRE NO	TO	REMARK
1	MG300	LTEB	3GKW-70
2	MG343	TM2	MAKER SUPPLY

EARTH BRUSH 3

TERMINAL NO	WIRE NO	TO	REMARK
1	MG302	HTEB	4GKW-AX-J-185
2	MBG351	BOGIE FRAME 2	4GKW-AX-35

EARTH BRUSH 4

TERMINAL NO	WIRE NO	TO	REMARK
1	MG303	HTEB	4GKW-AX-J-185
2	MG345	TM4	4GKW-AX-35

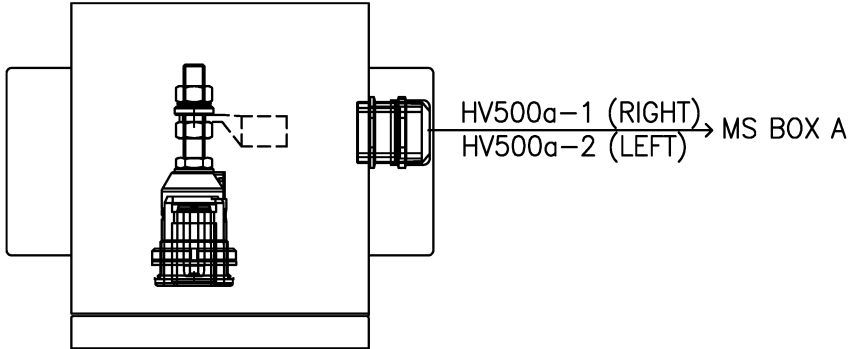
MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315–1000	1000–2000	2000–4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

STINGER BOX (RIGHT)

WIRE NO	TO	REMARK
HV500a-1	MS BOX A	4GKW-AX-185

STINGER BOX (LEFT)

WIRE NO	TO	REMARK
HV500a-2	MS BOX A	4GKW-AX-185



STINGER CONNECTOR WIRING

						PRODUCT	KMRCL RS(3R) METRO CARS					
						REF DRG						
						MATERIAL						
						HEAT TREAT.	APPD	KC SHAHSIKANTH		27.01.2018		
						SURFACE TREAT.	REVD	SADHASIVAM M		27.01.2018		
						TITLE	CHKD	V.SARANYA		27.01.2018		
						PIN ASSIGNMENT, EARTH BRUSH, STINGER - M CAR	DRWN	SUSHILA PATIL		27.01.2018		
							SCALE			SHEET	Wt.(Kg)	
							NTS			1 OF 1	0	
							DRG No.					ALT
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEML LIMITED NEW FRONTIERS. NEW DREAMS	942-38721					



BEML LIMITED

942-38721

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	VALUE	SYMBOL
ROUGHNESS	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50		

5

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1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)								QUALITY OF WELD JOINTS REF, RD 230 MEDIUM		
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

MC

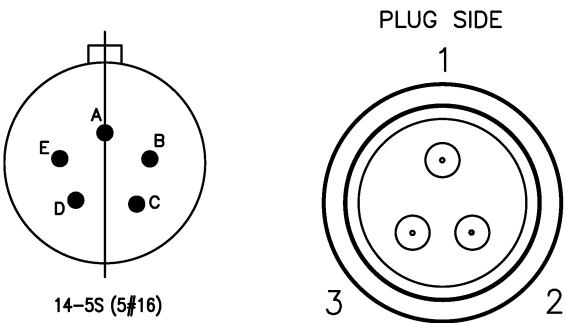
ANTI-SKID VALVE1 (HL6)

Pin No	ASV1	DMV PLUG SIDE	REMARKS
A	M4212/1-1	3	BECU T/B
B	M4212/2-1	1	BECU T/B
C	M4212/3-1	2	BECU T/B
D			
E			

ANTI-SKID VALVE2 (HL20)

Pin No	ASV2	DMV PLUG SIDE	REMARKS
A	M4213/1-1	3	BECU T/B
B	M4213/2-1	1	BECU T/B
C	M4213/3-1	2	BECU T/B
D			
E			

NOTE1: M4212/S-1,M4213/S-1 SHALL BE TERMINATION APPLIED.



* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE GKW EMC 3x1.0

COUPLER_JB

P/NO	X01		TO	REMARK
	Wire No	Car Side		
01	M2132		EDB	3GKW-1.5
02	M2133		EDB	
03				
04	M2133-1		EDB	
05	M2141		EDB	
06				
07				
08				

1 M2141 M2133-1

SP141
SP142

						PRODUCT	KMRCL RS(3R) METRO CARS					
						REF DRG						
						MATERIAL						
						HEAT TREAT.		APPD	KC SHAHSIKANTH	27.01.2018		
						SURFACE TREAT.		REVD	SADHASIVAM M	27.01.2018		
						TITLE	PIN ASSIGNMENT, ANTI SKID VALVE, COUPLER_JB - M CAR			CHKD	V.SARANYA	27.01.2018
DRWN	SUSHILA PATIL	27.01.2018										
SCALE		SHEET	Wt.(Kg)									
	Autocoupler pin no 4 & 5 interchanged.	DATE	24.03.2018	BY	V.SARANYA	CHKD	G.VISHWANATH	APPD	K.C.SHAHSIKANTH	NTS	1 OF 1	0
ALT.NO.	ECN NO/CHANGES									DRG No.	ALT	
							BEML LIMITED			942-38722		
						NEW FRONTIERS. NEW DREAMS						

BEML LIMITED
NEW FRONTIERS. NEW DREAMS

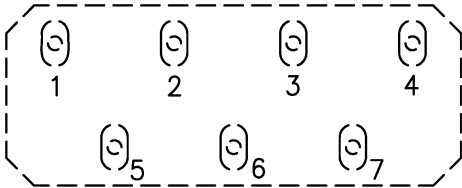
942-38722

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SURFACE ROUGHNESS													

TERMINAL NO	WIRE NO	TO	REMARK
1	WPE360a	HTJB	4GKW-AX-185
2	HV799-5	VVVF INVERTER	4GKW-AX-185
3	MG302	EARTH BRUSH 3	4GKW-AX-J-185
4	MG301	EARTH BRUSH 1	4GKW-AX-J-185
5	MG303	EARTH BRUSH 4	4GKW-AX-J-185
6	HV799-6	VVVF INVERTER	4GKW-AX-185
7			

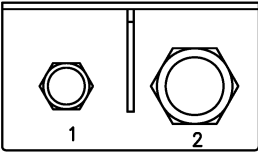


TERMINAL ARRANGEMENT

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

HIGH TENSION JUNCTION BOX

TERMINAL NO	WIRE NO	TO	REMARK
1	HV501-1	MS BOX B	4GKW-AX-70
2	WPE360a	HTEB	4GKW-AX-185



TERMINAL ARRANGEMEENT

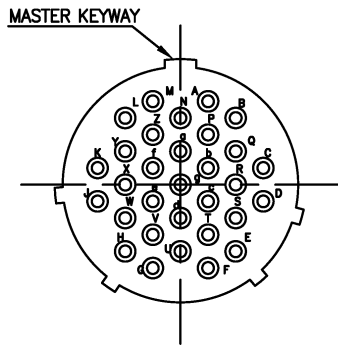
						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	27.01.2018	
						SURFACE TREAT.		REVD	SADHASIVAM.M	27.01.2018	
						TITLE	PIN ASSIGNMENT, HTEB, HTJB -M CAR			CHKD	V.SARANYA
DRWN	SUSHILA PATIL	27.01.2018									
SCALE			SHEET	Wt.(Kg)							
NTS			1 OF 1	0							
						DRG No.				ALT	
						942-38723					
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEML LIMITED NEW FRONTIERS. NEW DREAMS					

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	VALUE	SYMBOL
	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50		
SURFACE ROUGHNESS														

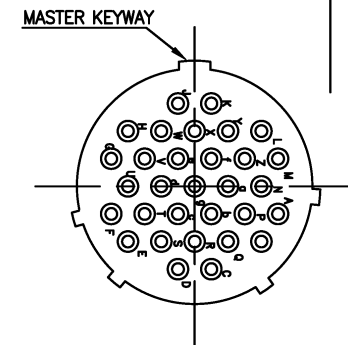
CN1

PIN NO	WIRE NO	TO	REMARK
A	1002	LTJB1	
B	1032-2	EDB	
C	1050-2	LTJB1	
D	1040-1	EDB	
E	1207-1	EDB	
F	1209-1	EDB	
G	1215	LTJB1	
H	1217-1	EDB	
J	1205-1	EDB	
K	1219	LTJB1	
L	2054	LTJB1	
M			
N			
P	2043	LTJB1	
Q	3003-2	EDB	
R	3004-2	EDB	
S	3005-2	EDB	
T	3006-2	EDB	
U	3010-2	EDB	
V	3016-2	EDB	
W	3011-2	EDB	
X	4026	BECU TB	
Y	4028-1	BECU TB	
Z	3031-2	EDB	
a	3013-2	EDB	
b	3018-2	EDB	
c	3019-2	EDB	
d	3060-1	EDB	
e	4062-1	EDB	
f	4070-1	EDB	
g	4103-1	EDB	



CN1 : 24-31S (31 #16)

SP123
SP124 - BECU T/B
SP125
D3-3- EDB

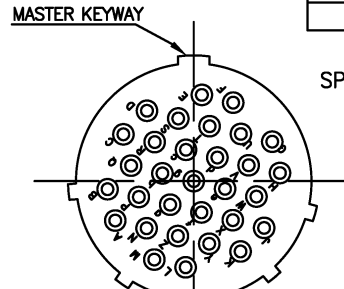


CN2 : 24-31SW (31 #16)

CN2

PIN NO	WIRE NO	TO	REMARK
A	4110-1	EDB	
B	4152-2	EDB	
C	4153-2	EDB	
D	4600-3	BECU T/B	
E	7071-1	EDB	
F	4123-3	BECU T/B	
G			
H			
J	5064	LTJB1	
K	6009-2	EDB	
L	6046-2	EDB	
M	6034-10	EDB	
N	6014-2	EDB	
P	6051-2	EDB	
Q	6073-1	EDB	
R	6076-1	EDB	
S	6112-2	EDB	
T	4155	PSB 2	
U			
V	5000	LTJB1	
W			
X			
Y			
Z	9214	LTJB1	
a	9215	LTJB1	
b	9216	LTJB1	
c	D4-3	EDB	
d	GRL1-3	EDB	
e	M3020a-1	BECU T/B	
f	2054C	LTJB1	
g	2054B	LTJB1	

SP120
SP121 - EDB
SP122
SP126 - PSB2
SP110
SP111 - LTJB1
SP112
SP135
SP136 - EDB
SP137



CN3 : 24-31SX (31 #16)

CN3

PIN NO	WIRE NO	TO	REMARK
A	SP001-1	EDB	
B	SP002-1	EDB	
C	SP003-1	EDB	
D	SP004-1	EDB	
E	SP005-1	EDB	
F	SP006-1	EDB	
G	SP007-1	EDB	
H	SP008-1	BECU T/B	
J	SP009-1	BECU T/B	
K	SP010-1	BECU T/B	
L	3085/1-2		
M	3085/2-2	BECU T/B	GKW EMC 2x1.0
N	3085/S-2		
P	SP026/1-1	BECU T/B	GKW EMC 2x1.0
Q	SP026/2-1		
R	SP027/1-1	BECU T/B	GKW EMC 2x1.0
S	SP027/2-1		
T	SP028/1-1	BECU T/B	GKW EMC 2x1.0
U	SP028/2-1		
V	SP029/1-1	BECU T/B	GKW EMC 2x1.0
W	SP029/2-1		
X	SP030/1-1	BECU T/B	GKW EMC 2x1.0
Y	SP030/2-1		
Z	SP031/1-1	BECU T/B	GKW EMC 2x1.0
a	SP031/2-1		
b			
c			
d			
e			
f			
g			

SP026/S-1, SP027/S-1, SP028/S-1, SP029/S-1, SP030/S-1, SP031/S-1, SHALL BE CONNECTED TO b PIN AFTER SLEEVE JOINT.

9244/1,2,S
9246/1,2,S
SP9076/1,2,S-AAU

* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 3GKW-1.5

△					PRODUCT	KMRCL RS(3R) METRO CARS			
△					REF DRG				
△					MATERIAL				
△					HEAT TREAT.		APPD	K.C.SHASHIKANTH	29.01.2017
△					SURFACE TREAT.		REVD	SADHASIVAM.M	29.01.2017
△					TITLE		CHKD	V.SARANYA	29.01.2017
△							DRWN	SUSHILA PATIL	29.01.2017
1	X5 & X10 Pin details updated	DATE	12.04.2018	BY	AMBIKA	CHKD	GIRISH	APPD	KCS
ALT.NO.	ECN NO/CHANGES	DATE	12.04.2018	BY	AMBIKA	CHKD	GIRISH	APPD	KCS
						PIN ASSIGNMENT, LTJB2 - M CAR		SCALE	NTS
						DRG No.		SHEET	1 OF 3
						ALT		Wt.(Kg)	0
						ALT		Wt.(Kg)	0

BEML LIMITED

942-38724

A3

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GRADE No.	N1	0.025
VALUE	N2	0.05
SYMBOL	N3	0.1
	N4	0.2
	N5	0.4
	N6	0.8
	N7	1.6
	N8	3.2
	N9	6.3
	N10	12.5
	N11	25
	N12	50
SURFACE ROUGHNESS		

5

4

3

2

1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 - 6	6 - 30	30 - 120	120 - 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)							QUALITY OF WELD JOINTS REF, RD 230 MEDIUM			
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.						STATUS:				
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

CN4

PIN NO	WIRE NO	TO	REMARK
A	SP032/1-1	BECU T/B	GKW EMC 2x1.0
B	SP032/2-1		
C	SP033/1-1		
D	SP033/2-1		
E	9240/1	LTJB1	GKW EMC 2x1.0
F	9240/2	LTJB1	GKW EMC 2x1.0
G	9241/1	LTJB1	GKW EMC 2x1.0
H	9241/2	LTJB1	GKW EMC 2x1.0
J	9242/1	LTJB1	GKW EMC 2x1.0
K	9242/2	LTJB1	GKW EMC 2x1.0
L	9243/1	LTJB1	GKW EMC 2x1.0
M	9243/2	LTJB1	GKW EMC 2x1.0
N	9245/1	LTJB1	GKW EMC 2x1.0
P	9245/2	LTJB1	GKW EMC 2x1.0
Q	9197	LTJB1	GKW EMC 2x1.0
R	9198	LTJB1	GKW EMC 2x1.0
S	9199	LTJB1	GKW EMC 2x1.0
T	9200	LTJB1	GKW EMC 2x1.0
U	9217	LTJB1	GKW EMC 2x1.0
V	9218	LTJB1	GKW EMC 2x1.0
W	9219	LTJB1	GKW EMC 2x1.0
X	9220	LTJB1	GKW EMC 2x1.0
Y	SP020/1-1	EDB	GKW EMC 2x1.0
Z	SP020/2-1	EDB	GKW EMC 2x1.0
a	SP020/S-1	EDB	GKW EMC 2x1.0
b	SP021/1-1	EDB	GKW EMC 2x1.0
c	SP021/2-1	EDB	GKW EMC 2x1.0
d	SP021/S-1	EDB	GKW EMC 2x1.0
e	SP022/1-1	EDB	GKW EMC 2x1.0
f	SP022/2-1	EDB	GKW EMC 2x1.0
g	SP022/S-1	EDB	GKW EMC 2x1.0

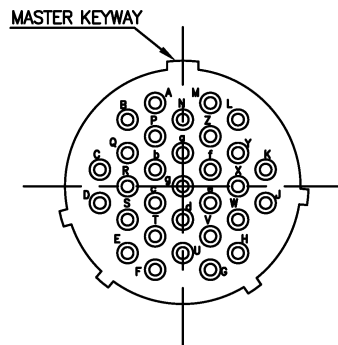
CN5

PIN NO	WIRE NO	TO	REMARK
A	SP023/1-1	EDB	GKW EMC 2x1.0
B	SP023/2-1		
C	SP023/S-1		
D	SP024/1-1	EDB	GKW EMC 2x1.0
E	SP024/2-1		
F	SP025/1-1	EDB	GKW EMC 2x1.0
G	SP025/2-1		
H			
J			
K			
L			
M			
N			
P			
Q			
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
f			
g			

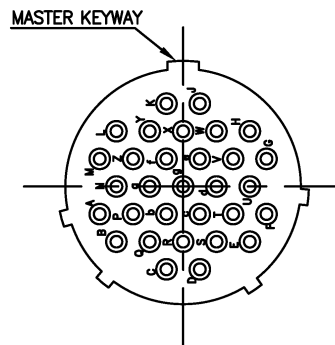
CN6

PIN NO	WIRE NO	TO	REMARK
A	SP011-1	BECU TB	
B	SP012-1		
C	SP013-1		
D	SP014-1		
E	SP015	LTJB1	
F			
G			
H			
J			
K			
L			
M			
N			
P			
Q			
R			
S			
T			
U			
V			
W			
X			
Y			
Z			
a			
b			
c			
d			
e			
f			
g			

SP032/S-1,SP033/S-1,9240/S,9241/S,9242/S,9243/S,9245/S,9197/S,9198/S,9199/S, 9200/S,9217/S,9218/S,9219/S,9220/S SHALL BE CONNECTED TO 'a' PIN AFTER SLEEVE JOINT. SP023/S-1,SP024/S-1,SP025/S-1 SHALL BE CONNECTED TO 'h' PIN AFTER SLEEVE JOINT.



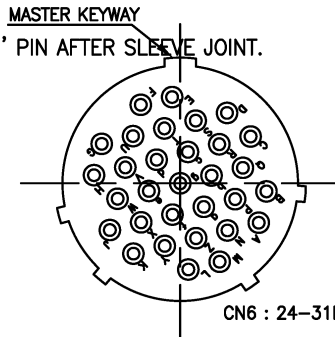
CN4 : 24-31P (31 #16)



CN5 : 24-31PW (31 #16)

SP034/1,2,S
SP035/1,2,S

LTJB1



CN6 : 24-31PX (31 #16)

SP100
SP101
SP102
SP103
SP104

BECU TB

* NOTE

1. UNLESS OTHERWISE SPECIFIED ALL CABLE SHALL BE THE 3GKW-1.5

△						PRODUCT	KMRCL RS(3R) METRO CARS			
						REF DRG				
△						MATERIAL				
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	29.01.2017
△						SURFACE TREAT.		REVD	SADHASIVAM.M	29.01.2017
						TITLE				
△						PIN ASSIGNMENT, LTJB2 - M CAR				
						CHKD	V.SARANYA		29.01.2017	
						DRWN	SUSHILA PATIL		29.01.2017	
						SCALE				
1	X5 & X10 Pin details updated	DATE	12.04.2018	BY	AMBIKA	CHKD	GIRISH	APPD	KCS	
ALT.NO.	ECN NO/CHANGES									
						b e m l BEML LIMITED				
						942-38724				
						DRG No.				ALT
										0

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2

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A3

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GRADE No.	N1	N2	N3	N4	N5	N6	N7	N8	N9	N10	N11	N12	SYMBOL
VALUE	0.025	0.05	0.1	0.2	0.4	0.8	1.6	3.2	6.3	12.5	25	50	
SURFACE ROUGHNESS													

5

4

3

2

1

MACHINING DEVIATIONS FOR LINEAR DIMENSIONS	RANGE	0 – 6	6 – 30	30 – 120	120 – 315	315-1000	1000-2000	2000-4000	ABOVE 4000	RA
	TOLERANCE	±0.1	±0.2	±0.3	±0.5	±0.8	±1.2	±2	±3	~
FOR DIMENSIONAL TOLERANCES OF SHEET METAL PARTS AND WELDED STRUCTURES, REFER STD. RD-227										
UNSPECIFIED TOLERANCE FOR LINEAR AND ANGULAR DIMENSIONS REF. IS 2102 (PT-1) (MEDIUM)								QUALITY OF WELD JOINTS REF, RD 230 MEDIUM		
VALUES OF SURFACE TEXTURE SHALL BE AS PER COMPANY STD DS. 1012.C.							STATUS:			
WELDING SHALL BE CARRIED OUT AS PER IS: 9595-96										

X5

PIN NO	WIRE NO	TO	REMARK
f1	8319/1	TCMS CN1	12584038 (RADOX RAILCAT CAT5e)
f2	8319/2		
f3	8319/3		
f4	8319/4		
e1	8320/1	TCMS CN2	12584038 (RADOX RAILCAT CAT5e)
e2	8320/2		
e3	8320/3		
e4	8320/4		
d1	9558/1	LTJB1	1767-4621 (SAB CAT 7A R)
d2	9558/2		
d3	9558/3		
d4	9558/4		
d5	9558/5		
d6	9558/6		
d7	9558/7		
d8	9558/8		
c1	9557/1	ETHERNET SWITCH	1767-4621 (SAB CAT 7A R)
c2	9557/2		
c3	9557/3		
c4	9557/4		
c5	9557/5		
c6	9557/6		
c7	9557/7		
c8	9557/8		
b1	SPCAN1-1/1	BECUTB	12552039 (RADOX DATABUS 120 OHM nx0.5 XMS)
b2	SPCAN1-1/2		
b3	SPCAN1-1/3		
a1	SPCAN2-1/1	BECUTB	
a2	SPCAN2-1/2		
a3	SPCAN2-1/3		

X10

PIN NO	WIRE NO	TO	REMARK
f1	9255/1	L3 SWITCH_2	12584038 (RADOX RAILCAT CAT5e)
f2	9255/2		
f3	9255/3		
f4	9255/4		
e1	9256/1	L3 SWITCH_2	12584038 (RADOX RAILCAT CAT5e)
e2	9256/2		
e3	9256/3		
e4	9256/4		
d1	9257/1	L3 SWITCH_2	12584038 (RADOX RAILCAT CAT5e)
d2	9257/2		
d3	9257/3		
d4	9257/4		
c1	9233/1	L3 SWITCH_1	12584038 (RADOX RAILCAT CAT5e)
c2	9233/2		
c3	9233/3		
c4	9233/4		
b1	9234/1	L3 SWITCH_1	12584038 (RADOX RAILCAT CAT5e)
b2	9234/2		
b3	9234/3		
b4	9234/4		
a1	9235/1	L3 SWITCH_1	12584038 (RADOX RAILCAT CAT5e)
a2	9235/2		
a3	9235/3		
a4	9235/4		

X10 TOP

CODE PIN:MALE

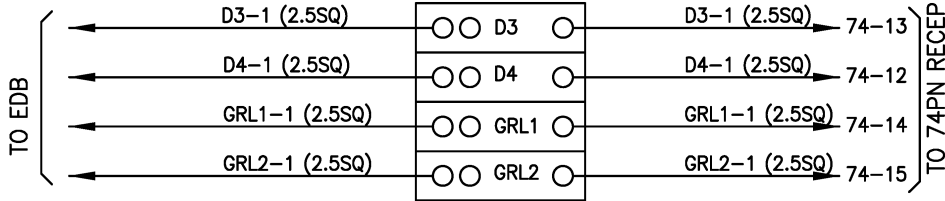
Module a, b

- 1-TX+ (YELLOW)
2-RX+ (WHITE)
3-TX- (ORANGE)
4-RX- (BLUE)

CODE PIN:FEMALE

BOTTOM

T/B WIRING



X5-C/D MODULE FOR CCTV

BROWN/WHITE

7

8

BROWN

CODE PIN:MALE

BLUE/WHITE

5

4

BLUE

ORANGE/WHITE

3

6

ORANGE

CODE PIN:FEMALE

GREEN/WHITE

1

2

GREEN

FRONT

X5 TOP

CODE PIN:MALE

Module a, b

- 1-TX+ (YELLOW)
2-RX+ (WHITE)
3-TX- (ORANGE)
4-RX- (BLUE)

BOTTOM

						PRODUCT	KMRCL RS(3R) METRO CARS				
						REF DRG					
						MATERIAL					
						HEAT TREAT.		APPD	K.C.SHASHIKANTH	29.01.2018	
						SURFACE TREAT.		REVD	SADHASIVAM.M	29.01.2018	
						TITLE		CHKD	V.SARANYA	29.01.2018	
						PIN ASSIGNMENT, LTJB2 - M CAR		DRWN	SUSHILA PATIL	29.01.2018	
								SCALE		SHEET	Wt.(Kg)
	X5 & X10 Pin details updated	12.04.2018	AMBIKA	GIRISH	KCS			NTS		3 OF 3	0
ALT.NO.	ECN NO/CHANGES	DATE	BY	CHKD	APPD	BEM LIMITED NEW FRONTIERS. NEW DREAMS	DRG No.				ALT
							942-38724				

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4

3

2

1

A3